

Final

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Github Repository

Set Up

```
library(tidyverse)
library(here)
library(janitor)

nests <- read.csv(here("data", "nest_data_final.csv"))
```

Problems

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, they used a binomial “log odds” regression to relate and predict (*Recurvirostra americana*) habitat occupation to distance from water edge.

In part 2, they used a Chi-Squared test to compare expected vs observed counts for habitat type and occupation by species.

b. Suggestions for Rewriting

Utilizing binomial logistic regression analysis, we found a significant relationship between distance from the water’s edge and American Avocet (*Recurvirostra americana*) microhabitat use.

Utilizing a Chi-Squared test, we found a significant relationship between bird species (American Avocet, Forster’s Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further Analysis

One additional variable that could influence the probability of American Avocet (*Recurvirostra americana*) microhabitat use is **salinity**. This variable is continuous, numerical (ppm) and often calculated after measuring electrical conductivity in micro Siemens per centimeter ($\mu\text{S}/\text{cm}$). Salinity influences a plethora of biological interactions, specifically dehydration, [bodyweight](#) and [feeding/preening behavior in American Avocet chicks](#).

Problem 2. Data Analysis

a. Clean & Wrangle

```
nests_clean <- nests |>
  select(case_control, ant_species, height_cm) |>
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  mutate(ant_name = case_match(
    ant_species,
    "AB" ~ "C.sjostedti",
    "RRB" ~ "C.mimosae",
    "BBR" ~ "C.nigriceps"
  )) |>
  mutate(ant_name = as_factor(ant_name),
         ant_name = fct_relevel(ant_name,
                                "C.sjostedti",
                                "C.mimosae",
                                "C.nigriceps"))

str(nests_clean)
```

```
'data.frame': 270 obs. of 4 variables:
 $ case_control: int 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species : chr "RRB" "RRB" "RRB" "RRB" ...
 $ height_cm : num 392 310 513 154 324 ...
 $ ant_name : Factor w/ 3 levels "C.sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...
```

```
slice_sample(nests_clean, n = 10)
```

```
  case_control ant_species height_cm ant_name
1            1          BBR    157.5 C.nigriceps
```

2	0	RRB	166.5	C.mimosae
3	0	RRB	236.0	C.mimosae
4	0	RRB	157.0	C.mimosae
5	0	RRB	207.0	C.mimosae
6	0	BBR	140.0	C.nigriceps
7	1	RRB	249.5	C.mimosae
8	1	BBR	234.5	C.nigriceps
9	0	RRB	241.5	C.mimosae
10	0	RRB	138.5	C.mimosae

b. Response Variable

The 1s and 0s present within the “case_control” column of the data represent if a bird nest was observed (1) or not (0) within a specific tree.

c. Purpose of Study

Tree height is a continuous, numerical variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because the tree contains more space for nesting and is further away from the ground and therefore potential predators. (Section 3: Results, ¶2)

Acacia ant species is a categorical, nominal variable. The four studied ant species differ in aggression of protecting their host trees, as aggressive protection increases, bird nesting also increases because of this inherent nest protection. (Section 3: Results, ¶3)

d. Table of Models

Model number	Ant Species	Height	Nests	Predictor List
0				no predictors (null model)
1	X	X	X	all predictors (full model)
2	X	X		Ant Species and Height
3	X		X	Ant Species and Nests
4		X	X	Height and Nests
5	X			Ant Species
6		X		Height
7			X	Nests

Model number	Ant Species	Height	Predictor List
0			null model
1	X	X	no interaction term
2	X	X	interaction term

e. Running the Models

```
model0 <- lm(case_control ~ 1,
             data = nests_clean)
model0
```

Call:

```
lm(formula = case_control ~ 1, data = nests_clean)
```

Coefficients:

```
(Intercept)
      0.2222
```

- f. Selecting the Best Model**
- g. Checking Diagnostics**
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- k. Interpreting Results**

Problem 3. Affective & Exploratory Visualizations

- a. Comparing Visualizations**
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